Exploratory Data Analysis of New York City Airbnb Listings (2024)

Welcome to the exploratory data analysis (EDA) project on **New York City Airbnb listings for 2024**. This project uncovers insights into pricing, availability, neighborhood trends, and room type distributions using Python and key data visualization libraries.

📌 Project Overview

The goal of this project is to:

- Understand the distribution of Airbnb listings across NYC boroughs.
- Analyze pricing patterns and room type availability.
- Identify correlations between different variables.
- Visualize key trends in availability, neighborhood popularity, and host behaviors.
- Clean and prepare the data for meaningful insights.

🧰 Tools & Technologies

- **Programming Language:** Python
- **Libraries Used:**
  - pandas – data manipulation & analysis
  - numpy – numerical computing
  - matplotlib & seaborn – data visualization

📂 Dataset

- **Source:** Inside Airbnb
- **Dataset:** NYC Airbnb listings (2024)
- **Records:** 20,000+ listings
- **Features Include:**

- name, host_name, neighbourhood, room_type, price, availability_365, etc.

🧼 Data Cleaning & Preprocessing

- Removed duplicates and null values
- Cleaned and standardized column formats
- Filtered extreme outliers in pricing
- Converted datatypes where necessary for analysis

📊 Key Insights

- **Manhattan** has the highest number of listings, especially in neighborhoods like Harlem and the Upper West Side.
- **Private rooms** are the second most common room type after entire homes/apartments.
- Prices vary significantly by **neighborhood and room type**.
- Listings with **high availability** tend to have more competitive prices.
- A few hosts dominate the platform with **multiple listings**.

📈 Visualizations

- **Price distribution** across neighborhoods and boroughs
- **Room type** distribution pie chart
- **Availability heatmaps**
- **Host analysis**: number of listings per host
- **Correlation heatmaps** for numerical variables

💡 Future Improvements

- Incorporate **text analysis** on listing names and reviews
- Apply **machine learning** for price prediction
- Compare **year-over-year trends** using previous datasets